Cardiovascular System (CARD) · NPLEX-P1-CARD-01

CV-1 Cardiovascular Embryology

Work through each activity on blank paper with the notes chapter closed, then open the answer key and mark what you missed. Do the two brain dumps for a competency on different days rather than back to back.

Work each activity without your notes first. Rate your confidence before you check. Then read the notes, write down what you missed, and set a date to come back to it.

Competency tracker

Competency	BD1 done	BD2 done	Confidence 1 to 4	Next review date
NPLEX-P1-CARD-01-01-C01				
NPLEX-P1-CARD-01-01-C02				
NPLEX-P1-CARD-01-01-C03				
NPLEX-P1-CARD-01-01-C04				
NPLEX-P1-CARD-01-01-C05				
NPLEX-P1-CARD-01-01-C06				
NPLEX-P1-CARD-01-01-C07				

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-01-01-C01-BD1

Draw the heart tube and name what each segment becomes

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C01 Activity: NPLEX-P1-CARD-01-01-C01-BD1 Date: _____

Without looking at your notes, draw the straight heart tube and label its five segments from caudal to cranial, then write beside each segment the main adult structure it becomes. Mark which segments come from the primary heart field and which come from the secondary heart field. Above the drawing write the sequence that builds the tube, from cardiac progenitors leaving the primitive streak to a tube that beats, and put the day or week beside each step. Name the three layers of the tube wall and write what each one gives rise to, including the tissue that supplies the epicardium and the coronary vessels. Finish by writing which parts of the atrial and ventricular walls end up smooth and which end up trabeculated, and explain that difference from the segment each wall came from.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-01-01-C01-BD2

A child with chronic sinusitis and an unusual chest film

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C01 Activity: NPLEX-P1-CARD-01-01-C01-BD2 Date: _____

A child has had a wet cough and recurrent ear and sinus infections since the newborn period, and the chest radiograph shows the cardiac apex on the right. Write which organelle is defective and which protein inside it is at fault, then explain step by step why that defect leaves the side of the heart unpredictable rather than always reversed. Write the proportion of these patients you would expect to have fully mirrored organs, the proportion with normal organ position, and the proportion with mixed sidedness, and name the triad that goes with the mirrored form. Next write why the same defect causes the cough, the ear disease, infertility in men, and a raised risk of ectopic pregnancy. Finish by writing which of these children carry a high risk of complex heart defects and which usually have a structurally normal heart, and state what the limb lead and precordial tracings look like when the whole body is mirrored.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-01-01-C01-RT01

Rebuild the left and right signalling sequence

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C01 Activity: NPLEX-P1-CARD-01-01-C01-RT01 Date: _____

Rebuild from memory the five-step sequence that runs from beating cilia at the primitive node to a heart tube that loops to the right, one step per line and in order. Beside the first step write where those cilia sit, how they are tilted, and which motor protein drives them. Beside the signalling steps write which genes switch on in left lateral plate mesoderm and which gene forms the midline barrier, and write what that barrier is for. Beside the last step write which organs take their sidedness from this signal. Underneath, write what happens to each of the five steps when the fluid flow at the node is absent, and draw the looping heart as it ends up, marking where the atria sit relative to the ventricles and which chamber ends up on the left.

Rebuild from memory: Left and right signalling that directs looping

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-01-01-C02-BD1

Draw atrial septation in four panels

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C02 Activity: NPLEX-P1-CARD-01-01-C02-BD1 Date: _____

Draw four panels of the developing atrial septum, from the first septum growing down to the sealed adult septum. In each panel label every septum and every opening present at that moment, and write the week the panel shows. Make clear in your drawing which opening is a hole inside the first septum and which opening is the gap left by the second septum, and write one sentence on how the two are different. Mark which septum is thin and which is thick, and which side of the heart each one sits on. Then write how the offset openings make a valve that passes blood one way only, state which way blood crosses before birth and why, and write what presses the valve shut after birth. Finish by naming the adult structure left behind and writing which part of it is the first septum and which part is the second.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-01-01-C02-BD2

An unsealed flap against a true hole

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C02 Activity: NPLEX-P1-CARD-01-01-C02-BD2 Date: _____

Two adults have an opening between the atria. One has a flap that never fused and the other has a genuine deficiency of septal tissue at the floor of the oval fossa. For each person write whether tissue is missing, which way blood crosses and why, what has to happen before any blood crosses at all, and what the usual consequence is. For the one with the true deficiency, write which chambers and vessels carry the extra volume and which chamber is spared, then write the murmur with its timing and where you hear it, explain why the flow through the hole itself is nearly silent, and write what the added rumble in diastole tells you about shunt size. Write what the limb lead axis and the shape of the complex in V1 look like, and explain why that pattern follows from the chamber that is loaded. Write out in four or five steps how decades of high pulmonary flow can reverse the shunt and produce blue skin. Finish by explaining why the second heart sound is both widely split and unchanging with breathing here, and why a block in the right bundle gives a wide split that still moves with breathing.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-01-01-C03-BD1

What the endocardial cushions build

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C03 Activity: NPLEX-P1-CARD-01-01-C03-BD1 Date: _____

Without your notes, draw the atrioventricular canal before and after the cushions fuse, and write where the canal sits before it shifts. List every structure the cushions go on to build, one per line, and beside each one write the week it is finished. Write the cell biology in order: which tissue signals, which signal families it uses, which cells change their character, what that change is called, and where those cells end up. Then write how the leaflets, the cords and the papillary muscles are carved out of cushion tissue, how many leaflets each orifice ends up with, and what tissue replaces the muscle inside the cords. Separately, write how the muscular part of the ventricular septum forms, how the membranous part closes, and name the three tissues that have to meet there. Finish by writing which part of the ventricular septum most defects sit in and explain that from the way it closes.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-01-01-C03-BD2

Why one cushion failure damages several structures

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C03 Activity: NPLEX-P1-CARD-01-01-C03-BD2 Date: _____

An infant with trisomy 21 is in heart failure at three months with a loud murmur, and the tracing shows a leftward and superior limb lead axis with a prolonged interval from the start of the P wave to the start of the complex. Write which developmental step failed, then list every structure that is abnormal because of that one failure. Write why this infant is in failure at three months while a child with a hole at the oval fossa of similar size can stay well for decades, using the pressure as well as the flow in your answer. Explain the limb lead axis and the lengthened interval from where the conducting tissue has been pushed and what has happened to the left anterior fascicle, and contrast that axis with the axis you expect at the oval fossa. Then write the difference between the complete and the partial form of this lesion in terms of the openings present, the valve leaflets, and the valve leak. Finish by writing what goes wrong with the leaflets in Ebstein anomaly and what blood has to do instead when the right atrioventricular orifice never opens.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Comparison table NPLEX-P1-CARD-01-01-C03-CP01

Three openings between the atria compared

Fill in every cell from memory, then check it against the notes and correct anything you missed.

Feature	Patent foramen ovale	Secundum atrial septal defect	Complete atrioventricular septal defect
Developmental step that failed			
Is septal tissue actually missing			
Where the opening sits			
State of the atrioventricular valves			
Direction of flow and what makes it move			
Limb lead axis on the tracing			
Age at which it usually causes trouble, and why			
Chromosomal or single gene association from the notes			

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-01-01-C04-BD1

Draw the spiral outflow septum and the three defects

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C04 Activity: NPLEX-P1-CARD-01-01-C04-BD1 Date: _____

Draw the single outflow tract in cross section at two levels and show the ridges that divide it, naming which ridge is superior and which is inferior and which way each one grows. Mark how far the septum twists and write what that twist achieves for the final position of each great vessel, including which vessel ends up anterior and which ends up posterior and which ventricle each one leaves. Write where the cells that build this septum come from, which arches they travel through, and the other structures in the neck they help to form. Then draw three more outflow tracts: one where the ridges never form, one where the septum comes down straight instead of spiralling, and one where the septum is pushed forward and upward. Beside each drawing name the defect and list its anatomical features. Finish by writing why displacing the outlet septum also leaves a hole in the ventricular septum.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-01-01-C04-BD2

Two blue newborns from two different failures

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C04 Activity: NPLEX-P1-CARD-01-01-C04-BD2 Date: _____

One newborn is deeply blue within hours of birth and gets worse on day two, with the great vessels lying one in front of the other on the film. A second child is pink for the first year and then has spells of deep blue colour during crying, with a heart on the film that has a raised apex and a scooped out upper left border. For each child name the defect, name the step of outflow division that failed, and write where the blood is going. For the first child write why the two circuits cannot supply each other, name the three places mixing can happen, and write what changed on day two. For the second child write why the colour depends on the balance between the resistance at the right ventricular outflow and the resistance in the body, and explain why mild obstruction leaves the child pink. Write where the murmur comes from and what happens to its loudness during a spell, and explain that from the flow. Finish by writing the two blood findings that follow years of low oxygen, and name the deletion syndrome that links outflow defects to a small thymus, low calcium and facial changes, with the gene involved and the reason the defect happens.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Prediction exercise NPLEX-P1-CARD-01-01-C04-PR01

Predict what happens during a spell and during squatting

A two-year-old with pulmonary outflow narrowing, a large malalignment hole in the ventricular septum, an overriding aorta and a thick right ventricle begins crying hard and turns deep blue. Predict each item below, write the direction of the change, and give the reason in one line from pressures, resistances or flow.

What changes	Up, down, or no change	Why, in one or two lines
What happens to the resistance in the systemic circulation while the child cries		
What happens to the resistance at the right ventricular outflow compared with the systemic resistance		
Which way blood crosses the hole in the ventricular septum, and what happens to the amount crossing		
What happens to the blood flow reaching the lungs and to the arterial oxygen saturation		
What happens to the loudness of the murmur, and from which part of the heart that murmur arises		
What the falling oxygen and rising acid in the blood do to the systemic resistance next, and what that does to the spell		
What happens to each of the items above when the child squats and folds the hips and knees		

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-01-01-C05-BD1

Draw the fetal circulation with its saturations

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C05 Activity: NPLEX-P1-CARD-01-01-C05-BD1 Date: _____

Draw the fetal circulation from the placenta back to the placenta. Label the umbilical vein, the three shunts, both atria and ventricles, the pulmonary trunk, the ascending and descending aorta, and the umbilical arteries, and mark the direction of flow at every point. Write the approximate oxygen saturation in the umbilical vein and in the umbilical arteries, and write the partial pressure of oxygen in the umbilical vein. Mark the ridge in the right atrium that aims the incoming stream at the atrial opening and write what it does. Then write the four reasons the resistance in the fetal lungs is high and the one reason the resistance in the body is low, and write which pressure is higher, the pulmonary trunk or the descending aorta. Finish by writing which ventricle handles the larger share of the combined output and what that explains about the thickness of the two ventricular walls at birth.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-01-01-C05-BD2

Why the fetal brain is well supplied at a low oxygen tension

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C05 Activity: NPLEX-P1-CARD-01-01-C05-BD2 Date: _____

The fetus lives with an arterial oxygen tension far below an adult value and still oxygenates its tissues. Write the four reasons that works, and for the haemoglobin reason name the two chains in fetal haemoglobin, the exact amino acid substitution responsible, the molecule that no longer binds, and the two half saturation pressures so the difference is on the page as numbers. Write what the double effect of carbon dioxide at the placenta does to the affinity on each side of the barrier and which way oxygen therefore moves. Next write out the route by which the best oxygenated blood reaches the heart muscle and the brain, naming the branches that leave the aorta before the mixing point, and write what reaches the lower body instead and why. Then predict and explain what happens to each ventricle and to the lower body if the vessel joining the pulmonary trunk to the descending aorta narrows in mid gestation, and what happens to the growth of the left heart if the atrial opening narrows instead. Finish by writing why embryonic chambers grow to the size they do.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Retrieval exercise NPLEX-P1-CARD-01-01-C05-RT02

Rebuild the path of blood through the fetus

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C05 Activity: NPLEX-P1-CARD-01-01-C05-RT02 Date: _____

Rebuild from memory the seven-step path of blood through the fetal circulation, one step per line and in order from the umbilical vein to the umbilical arteries. Beside each step write the saturation or the pressure relationship that explains why blood goes that way rather than the other way. Mark the two steps where a shunt is being used and write what each shunt bypasses. Beside the step through the liver, write what regulates how much blood takes the bypass and what that protects the heart from. Underneath the list, write in one line which great vein's blood ends up in the left heart and the brain and which great vein's blood ends up in the lower body, and name the vessel the umbilical arteries branch from.

Rebuild from memory: Path of blood through the fetal circulation

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-01-01-C06-BD1

Rebuild the changes at birth and the remnants table

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C06 Activity: NPLEX-P1-CARD-01-01-C06-BD1 Date: _____

Write the six-step sequence that turns a parallel circulation into a circulation in series, one step per line and in order, starting with the first breaths. Beside the breathing step write the three reasons the resistance in the lungs falls and name the two substances released by oxygen and shear stress. Beside the cord clamping step write what happens to the resistance in the body and to the venous return from below. Then write which pressure change closes the atrial opening, how long functional closure takes, and how long full anatomical fusion takes. Write the mechanism that closes the vessel between the pulmonary trunk and the aorta in two parts: the oxygen part, naming the channel that closes, the change in membrane voltage and the channel that then opens, and the prostaglandin part, naming the substance, where it came from, where it is now broken down, and the second messenger it worked through. Below that, build the whole table of fetal vessels and shunts with one row each and columns for the route and the reason for flow and for the adult remnant and its location, and include the cord structure that was never a vessel.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-01-01-C06-BD2

A vessel that stays open and the lesions that depend on it

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C06 Activity: NPLEX-P1-CARD-01-01-C06-BD2 Date: _____

Write why the vessel between the pulmonary trunk and the aorta stays open most often in babies born early, naming the two responses of immature ductal muscle, and name the infection and the environmental exposure that also make it likely. Then write the murmur it produces: its quality, its timing across the cardiac cycle, where you hear it, and why the sound continues rather than stopping. Write what happens to the pulse pressure and why, and which side of the heart carries the extra volume. Next write the three groups of lesions that become dangerous the moment this vessel closes, and for each one write whether the baby depends on it for mixing, for lung blood flow, or for body blood flow. Then explain how a large opening of this kind can end up making the toes blue and clubbed while the fingers stay pink, using the point at which the vessel joins the aorta in your answer, and write why a reversed shunt between the ventricles or between the atria colours fingers and toes alike. Finish by writing what is happening in a newborn whose lung resistance never falls despite normal heart anatomy, and where blood crosses in that baby.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Clinical application NPLEX-P1-CARD-01-01-C06-CA01

A newborn who collapses on the third day

A term newborn girl was well at delivery and fed normally for the first two days. On the third day she becomes pale, mottled and breathless. Temperature is 36.5 degrees Celsius (97.7 degrees Fahrenheit), pulse is 184 per minute, respirations are 68 per minute, and blood pressure in the right arm is 74 over 48 mm Hg. The femoral pulses cannot be felt and the liver edge is 3 cm below the costal margin. Oxygen saturation measured on the right hand is 96 percent and on the right foot is 82 percent. The chest radiograph shows a large heart and congested lung fields. Umbilical venous blood drawn at delivery had been reported as normal, and the newborn examination on day one, including femoral pulses, had been recorded as normal.

1. Recognize. Name the fetal structure that has just changed state and state what was supplying the lower body until the third day.

2. Explain. Explain from the fetal circulation why this baby was well on day one and why the signs appeared on day three, including why the liver enlarged and why the breathing is fast.

3. Differentiate. Write how this picture differs from a newborn who is blue in the first hours with parallel circuits, and from a newborn whose lung resistance never fell, naming the findings that separate the three.

4. Investigate. Write which measurements and which parts of the examination you would use to confirm where the obstruction sits, and what value or finding each one would give in this baby.

5. Decide. State which two signals normally close that structure and which of them was lost first in this baby, and state what the lower body circulation now depends on.

6. Justify. Justify the difference between the saturation at the right hand and the saturation at the right foot from the point at which that fetal vessel joins the aorta and the branches that leave the arch before it.

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 1 · foundational retrieval NPLEX-P1-CARD-01-01-C07-BD1

Build the arch artery table and draw both nerve loops

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C07 Activity: NPLEX-P1-CARD-01-01-C07-BD1 Date: _____

Draw the pharyngeal arch arteries running between the aortic sac and the paired dorsal aortae, number the arches, and write whether all of them are ever present at the same time. Then build a table from memory with one row for each embryonic source and a column for the adult derivative, covering the outflow trunk, the aortic sac on each side, the first and second arch arteries, the third, the fourth on the left and on the right, the fifth, the sixth on both sides, the left seventh intersegmental artery and the left dorsal aorta. Mark on your drawing the two segments of dorsal aorta that disappear. Underneath, draw both recurrent laryngeal nerves. Write which arch each nerve first hooked under, which vessel each ends up under in the adult, and why the two differ. Write which branch of the vagus supplies the fourth arch and which muscle it serves, and which muscles the recurrent nerve serves.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Brain dump 2 · integration and application NPLEX-P1-CARD-01-01-C07-BD2

Four remodelling errors and a hoarse voice

NPLEX Part: Part I Subject: Cardiovascular System (CARD) Chapter and topic: CV-1 Cardiovascular Embryology Competency: NPLEX-P1-CARD-01-01-C07 Activity: NPLEX-P1-CARD-01-01-C07-BD2 Date: _____

For each of the four situations below write which arch artery or dorsal aortic segment persisted or regressed abnormally, what the resulting anatomy is, and what the person notices: the right subclavian artery arising as the last branch of the arch and running behind the oesophagus; a vascular ring around the trachea and oesophagus; an interrupted arch between the left common carotid and the left subclavian; and a right laryngeal nerve that does not loop at all. For the interrupted arch write what the lower body depends on and explain why this one has the strongest link with the deletion syndrome in the notes. Then write which single muscle the recurrent nerve supplies that opens the vocal fold, where the fold comes to rest when the nerve is injured, and what the voice sounds like. Finish with the vulnerability of each nerve: list the structures along the left nerve's course in the chest and the conditions along it that cause hoarseness, name the dilated chamber and vessel that can trap it, list where the right nerve is at risk and from what, and name the operation that puts both at risk.

My response

Confidence before checking

☐ 1 Guessing ☐ 2 Developing ☐ 3 Nearly competent ☐ 4 Competent

After checking my notes

What did I remember correctly?

What did I forget or misunderstand?

What needs more practice?

Next retrieval date: _____

Board-style question NPLEX-P1-CARD-01-01-C02-Q01

An atrial opening that seals too early

Serial ultrasound examinations of a fetus at 24 weeks of gestation show that the flap valve between the atria has sealed against the thicker right sided septum and no flow crosses it. Follow up studies over the next 8 weeks show a progressively small left atrium and a small, thick walled left ventricle, with a right atrium and right ventricle that are larger than expected. Which of the following best explains the appearance of the left side of this heart?

A. Oxygenated blood is diverted into the pulmonary arteries, so the pulmonary vascular bed carries the volume instead

B. Pressure in the left atrium rises above pressure in the right atrium, so the left ventricle ejects against a greater load

C. Most left sided filling normally arrives across that opening, and chambers grow in proportion to the volume that passes through them

D. The endocardial cushions fail to fuse once flow through the atrioventricular canal is reduced, so the mitral orifice does not form

Why did you pick it?

Board-style question NPLEX-P1-CARD-01-01-C07-Q02

Hoarseness with a dilated aortic arch

A 68-year-old man comes to the physician because of 4 months of progressive hoarseness. Laryngoscopy shows that the left vocal fold lies near the midline and does not move on attempted abduction. Computed tomography of the chest shows a 6 cm aneurysm of the aortic arch. The nerve responsible for his hoarseness is held in the chest by a remnant of which of the following embryonic structures?

A. The distal part of the left sixth pharyngeal arch artery

B. The left fourth pharyngeal arch artery

C. The left seventh intersegmental artery

D. The left horn of the aortic sac

Why did you pick it?